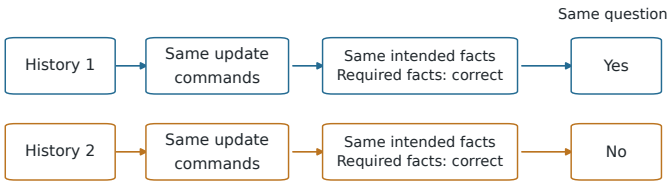
matched_history

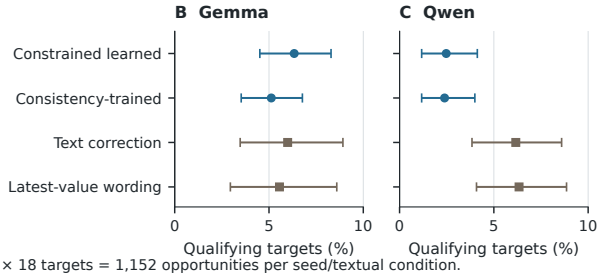
Correct current facts can leave history-dependent answers

Prospective matched-history test · Gemma and Qwen · all eight conditions



Two illustrated lanes; all 8 histories are tested. Answers shown are conceptual.

Equal correct categories do not imply equal probabilities or hidden states.

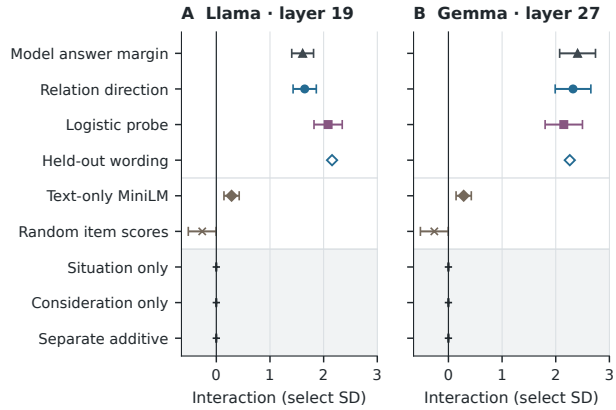


64 cases × 18 targets = 1,152 opportunities per seed/textual condition.
All scheduled targets remain in the denominator, including ineligible ones.
99.375% case-bootstrap intervals; learned seeds averaged within each case.
Qualifies: required facts correct in all histories; both references correct;
valid downstream answers differ. Target rate, not case prevalence or error rate.

construct_validity

A readable relation signal, with controls kept in view

Development evidence · 125 evaluation checkerboards · no strict-test claims



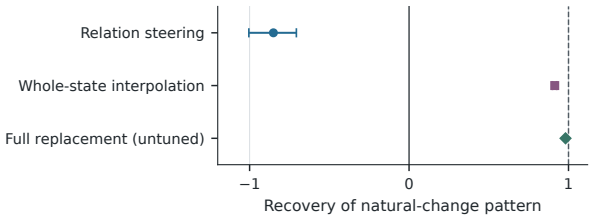
95% normal dyadic-robust intervals; open diamonds: no retained interval.
Bottom three controls cancel algebraically; random items are not directions.
Layer choice: separate select split. Held-out wording is still development.
Factual controls and mapping-reversal limits: see the truth-control supplement.

consequence_qualification

Matching the direct margin is not consequence recovery

Qualification-limited Gemma comparison · 96 contexts · not accuracy

Both tuned arms: 100% within the requested direct-margin tolerance.

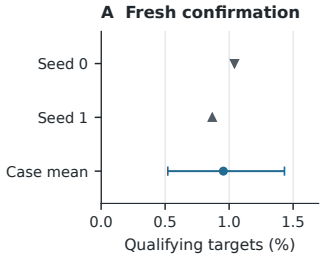


0 = unchanged state; 1 = natural change; negative = farther from that pattern.
95% context-bootstrap intervals. Full replacement is not margin-tuned.
Strong-target gates failed; matched direction unqualified (frozen direction used).
Llama natural-reference qualification failed. This is not a confirmatory success.
Whole-state: 0.91358 [0.89946, 0.92597]. Full replacement interval below:
0.98127 [0.97992, 0.98255] (narrower than its marker).

fresh_confirmation_utility

Fresh confirmation does not establish a utility advantage

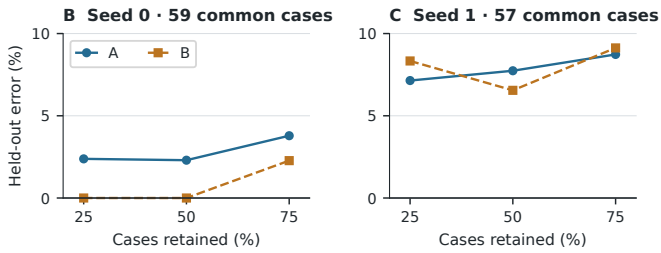
Gemma · one learned-update family · fresh 64-case cohort



0.95% mean
95% interval [0.52, 1.43]%

14/64 contributing cases
22 target/seed observations

1,152 scheduled coded targets per seed; ineligible targets remain.
Full factual/reference/logical + history-matching criterion.

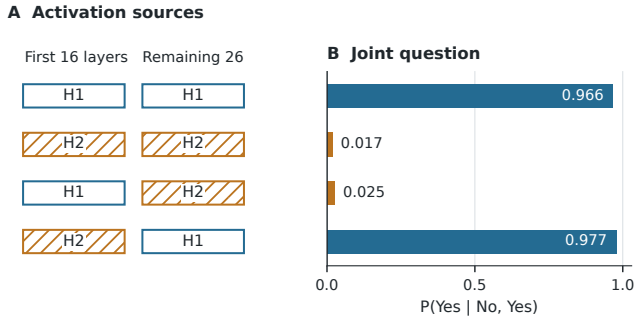


A: 6 checks at 1 history. B: 3 checks across 2 histories. Literal No/Yes.
Named 50%-retention error contrast B − A: −1.74 pp [−3.53, +0.65].
95% paired-case interval includes zero: superiority is not established.
Equal query counts do not establish equal GPU cost. Seeds are repeats.

selected_crossover

Changing the later activation source changes the answer

Selected Gemma example SSC1-FINAL-0026 · correct joint answer: Yes



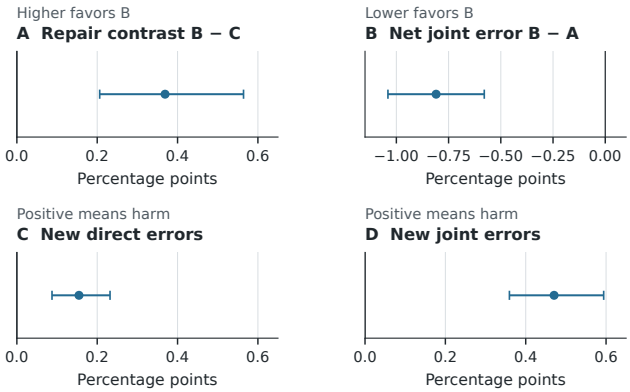
Early | later source	Dion direct P(correct)	Orla direct P(correct)
A | A	0.996255	0.999909
B | B	0.996415	0.999066
A | B	0.995544	0.998243
B | A	0.996973	0.999892

A = H1; B = H2. Color/hatching identify the later source, not correctness.
Combined before the question; weights fixed. One selected case, not prevalence.
Six selected cases: main format 5 later / 1 mixed; alternate 1 later / 2 mixed /
3 insufficient separation. Full six-case table and two-format extension retained.

repair_benefit_damage

Partial repair comes with newly introduced errors

Prospective Gemma population · all 256 cases · both stored seeds

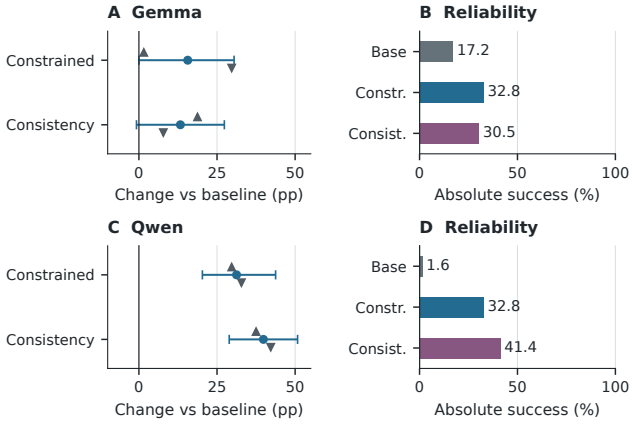


B: values 16–23 refreshed; C: values 24–31; A: unchanged computation.
98.75% paired-case intervals; 10,000 saved draws; all scheduled denominators.
Baseline W: 83 target/format/seed rows in 63 cases. B repairs 36 rows / 28 cases.
Strong-repair thresholds satisfied: FALSE (36/83 < 50%).
Joint errors: 3,490 → 2,893 / 73,728; full reference: 2,000 / 73,728.
Both partial refreshes require that full reference; no cost advantage established.

update_reliability

Improvement is not complete downstream reliability

All 34 questions correct under all four histories · 64 cases per model

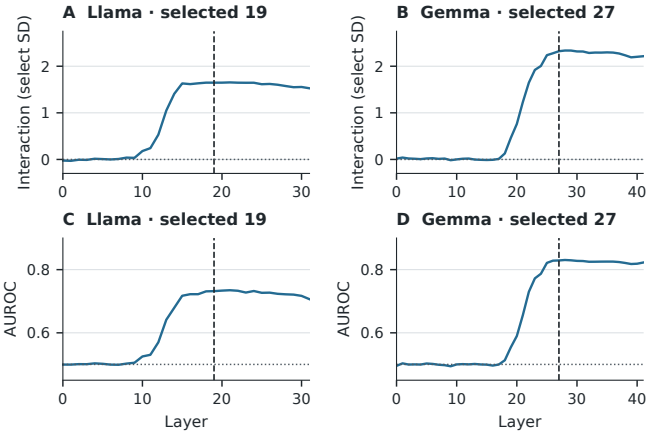


Dots: seed mean with saved 98.75% paired-case intervals. Triangles: seeds 0/1.
Baseline: attempted fact refresh + textual correction; fallback in 192/256 contexts.
Neither learned method reaches 50% mean complete-case success. Models stay separate.

layer_development

Layer emergence is development evidence

Selected layers are fixed by the select split, not by peaks in these curves

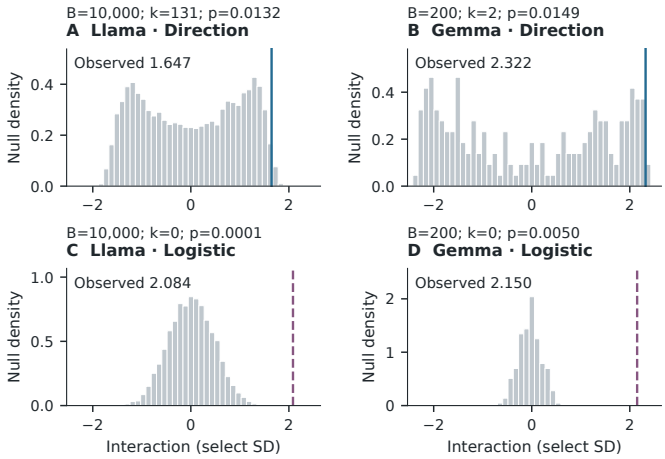


Mirrored-pair selection: 300 rows / 75 boards; evaluation: 500 rows / 125 boards.
Llama layer 19; Gemma layer 27. No smoothing or invented uncertainty bands.
These curves did not select layers and do not use the strict test.

permutation_nulls

All four permutation tests, with their original nulls

Situation-level label flips; probes refitted in the original saved analysis

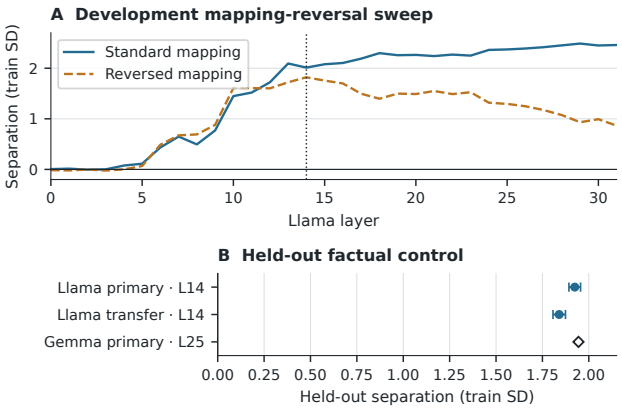


One-sided p = (k + 1)/(B + 1); no zero p-values or fitted null distributions.
Llama B=10,000; Gemma B=200. Original 48-bin shared histogram grid.
The earlier Llama B=200 comparator-improvement test is a different estimand.

truth_control

Truth control: distinguish selection from held-out testing

Neutral answer symbols swap meanings; later mapping sensitivity remains

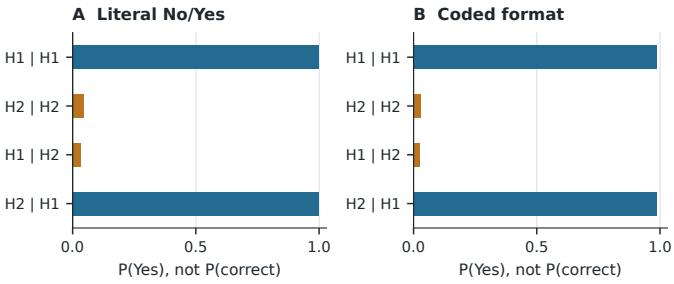


Llama: saved 95% group-bootstrap intervals, 2,000 replicates.
Gemma: point estimate only; no interval was retained.
Earlier literal-label control failed; passing does not validate every layer.

crossover_extension

Two answer formats of one retrospective crossover case

SSC1-FINAL-0035 · Ada and Gita oppose Theater · correct answer: No

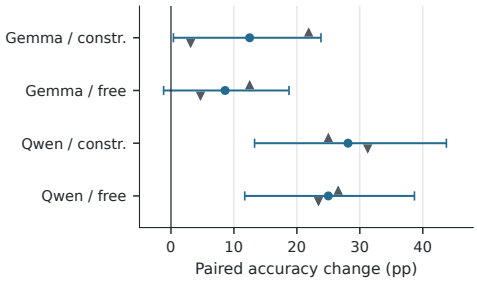


Early | later sources: first 16 / remaining 26 layers; weights fixed; pre-question.
Both histories: 2 fact changes, 517-token prefixes. Both formats are ONE case.
8 direct checks pass; min P(correct)=0.999230; max change=0.00008612.
Selected retrospectively, not a prevalence estimate; coded answers mapped to Yes.

editing_coverage

Broader single-edit training: bounded sequence gains

Gemma: 64 situations · Qwen: 32 situations · seeds remain repeated measurements

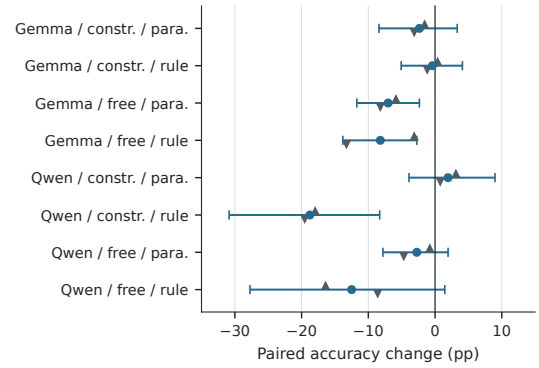


98.75% paired-situation intervals; triangles: separate seed effects.
Broader minus narrower training; all-question success after 2/3 updates.
Single-edit training only; these relative gains are not broad external generalization.

fixed_readers

Changing the reader does not establish an improvement

Gemma: 64 situations · Qwen: 32 situations · seeds remain repeated measurements

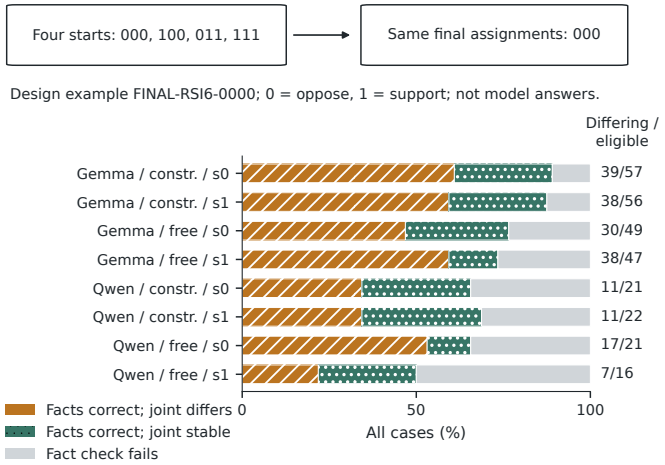


99.375% paired-situation intervals; original fixed early ordering; same-side query.
Paraphrase versus explicit rule are distinct readers on the SAME edited states.
Intervals crossing zero do not prove equivalence; both seeds are shown.

archival_source_history

Archival source-history counts are a different study

V6 descriptive census · original reader · four histories · no population intervals

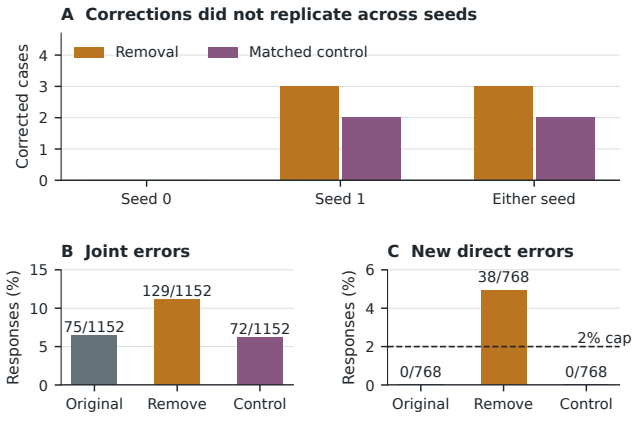


64 Gemma / 32 Qwen cases, shown separately for each editor and seed.
Stable joint answers may still be wrong. Right counts use fact-correct cases only.
Canonical replacement for BOTH S3 exports; not another replication of V10.

removal_screen

Message removal failed the historical safety screen

16 exposed cases: 14 prior positives + 2 controls · descriptive, not prevalence



Required: ≥3 corrected cases, ≥1 per seed, ≥2 more than control, lower joint error, and ≤2% new direct errors. Four of five gates fail; branch CLOSED.
Removal: 67 new joint errors versus 13 corrections; 38/768 new direct errors.
No fresh final cases were run. No descriptive screen intervals are invented.

fixed_worked_example

The fixed example was selected by hash, not by effect

QBR1-FINAL-0225 · target d0-02 · correct answer Yes · common suffix 20 tokens

A Stored seed 0

	Unchanged A	Refresh B values 16–23	Refresh C values 24–31	Full current record D
History 0	0.9973	0.9973	0.9973	0.9973
History 1	0.9957	0.9961	0.9960	0.9973
History 2	0.9979	0.9984	0.9981	0.9973
History 3	0.9954	0.9971	0.9960	0.9973
History 4	0.9986	0.9987	0.9986	0.9973
History 5	0.9968	0.9973	0.9969	0.9973
History 6	0.9985	0.9988	0.9984	0.9973
History 7	0.9945	0.9965	0.9950	0.9973
Facts	96/96	96/96	96/96	96/96
Other logic	64/64	64/64	64/64	64/64

B Stored seed 1

	Unchanged A	Refresh B values 16–23	Refresh C values 24–31	Full current record D
History 0	0.9980	0.9980	0.9980	0.9980
History 1	0.9920	0.9940	0.9922	0.9980
History 2	0.9973	0.9978	0.9973	0.9980
History 3	0.9938	0.9942	0.9941	0.9980
History 4	0.9973	0.9971	0.9973	0.9980
History 5	0.9844	0.9859	0.9846	0.9980
History 6	0.9964	0.9965	0.9965	0.9980
History 7	0.9878	0.9876	0.9890	0.9980
Facts	95/96	96/96	95/96	96/96
Other logic	57/64	57/64	57/64	56/64

All 64 displayed target responses are valid and correct. Cells show P(correct).
The first No/Yes joint query in frozen order was retained, regardless of effect.
D reuses one seed-specific context: invariance is automatic, not an extra success.
Both refreshes require preparing D. Keys, other values, positions, prompts and weights remain fixed. Full-precision scores and control counts are exported.